

Recovering Missing IoT Sensor Data Using Graph Signal Processing

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Declaration of Academic Integrity

We declare that this project report contains a literature survey and original work by the undersigned candidates as part of their Bachelor of Engineering Degree.

All information in this document has been obtained and presented by academic rules and ethical conduct.

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Abstract

Wireless- IoT sensor networks are among the most promising technologies of the current era because of their small size, lower cost, and ease of deployment. With the increasing number of wireless sensors, the probability of generating missing data also rises. This incomplete data could lead to disastrous consequences if used for decision-making. There is rich literature dealing with this problem. However, most approaches show performance degradation when a sizable amount of data is lost. Inspired by the emerging field of graph signal processing, this report performs a new study of a Sobolev reconstruction algorithm in wireless sensor networks. Experimental comparisons on several publicly available datasets demonstrate that the algorithm surpasses multiple state-of-the-art techniques by a maximum margin of 54%. We further show that this algorithm consistently retrieves the missing data even during massive data loss situations.

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Chapter 1

Introduction

Research is formalized curiosity.

Zora Neale Hurston, 1942

1.1 Background

In the age of the Internet of things (IoT), we can find widespread use of large-scale monitoring systems [1–3]. Sensors play the most crucial role in those monitoring systems by collecting data from the environment. The sensor nodes interact with one another and optionally with a base station to collectively monitor the application environment and make "intelligent" decisions in response to the perceived state. As the number of sensors becomes increasingly large, risks of registering lost or corrupted data due to external interference or sensor malfunction may arise. Also, it may become difficult to replace those defective sensors on many occasions due to factors like their deployment in remote locations and others. Fig. 1.1 shows one such scenario. These factors significantly restrict the deployment of sensor networks in various application domains [4, 5]. To compensate for the lost/ damaged data, we need to fill them with relevant one [6]. This report discusses some popular data recovery techniques available to date and further shows how a recently developed method improves over those techniques.

An enormous amount of data has been generated from various networks, like sensors and biological and social networks. Analyzing those data using traditional approaches becomes challenging for their irregular and complex structures. Under those situations, graphs provide a robust framework to excavate the networks' structure and hidden working mechanisms. Graph Signal Processing (GSP) [7] is one such field developed by incorporating the concepts of graph

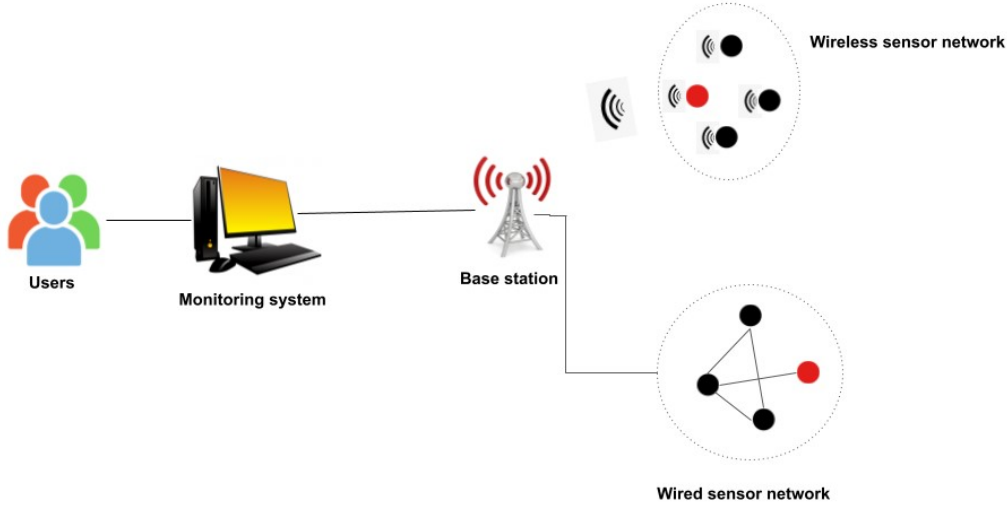


Figure 1.1 Sensor based monitoring system with missing data (A red circle denotes a failure sensor node)

theory and classical signal processing, which has proved itself to help solve multiple issues related to sensor networks [8]. In sensor networks, we usually represent the sensors as nodes of a graph and the sensor attributes (e.g., sensor readings like temperature, humidity, and others) as graph signals on nodes of the graph. Previously concepts based on GSP have been used in sensor networks, like in planning smart cities and intelligent grids [8], anomaly detection [9, 10], efficient sensor placement [11], and many others. However, there are only a very few instances in the literature where we can see the implementation of GSP for missing sensor data recovery. In this report, we discuss in detail how GSP-based methods can be useful for interpolating missing data in IoT sensor networks.

1.2 Literature Survey

We can find multiple approaches in the literature that deals with the problem of missing data in sensor networks. One such widely used method is the nearest-neighbour interpolation [12], which recovers the missing values by weighing and averaging the readings of the nearby sensors. Some statistical methods are also helpful for this task, like algorithms based on Expectation-Maximization(EM) [13]. Here the missing data is found by alternately computing the conditional expectation in E-step and updating the estimate by maximizing this conditional

expectation in the M-step. However, this method does not consider the spatio-temporal information in the data, thereby delivering sub-optimal performance in real-world situations. As an improvement upon this, a few of the recent approaches tackle the problem of spatio-temporal data recovery as a low-rank matrix completion (LRMC) problem, thereby estimating missing values by finding an appropriate low-rank approximation of the original incomplete data matrix. Nonetheless, these methods only capture the superficial features, failing to infer the embedded spatio-temporal dependencies in the sensor data. In the recent literature, some promising data-recovery methods are proposed, like the Probabilistic Matrix Factorization (PMF) [4]. In this method, the sensors are divided into different groups based on their similarity using k-means clustering, and then a PMF algorithm is applied within each group to recover the missing data. Even though this algorithm gives promising results, the performance of this algorithm is severely dependent upon the choice of hyperparameter k in k-means clustering and is prone to overfitting [14].

Sampling and recovery of graph signals is a crucial topic in graph signal processing theory and has led to some significant developments in numerous domains like climate monitoring, prediction of infectious diseases, etc. [15]. Recent approaches to sampling and recovering graph signals include extending the concept from static [16] to time-varying graph signals [17, 18]. Our work is inspired by one such time-varying graph signal reconstruction method, which is based on a Sobolev-norm recovery algorithm [15]. In our work [19], we formulate the problem of recovering missing sensor data as a reconstruction of time-varying graph signals. We show the efficiency of this recovery algorithm outperforms multiple state-of-the-art (SOTA) techniques [4, 12, 13, 20] even during massive data loss situations. To summarize our contributions:

- In this work [19], we introduce the concept of graph signal recovery for recovering missing data in wireless sensor networks.
- We show that the Sobolev norm reconstruction algorithm improves upon previous works that performed similar tasks.
- We compare our method against the SOTA approaches by testing our algorithm on two open-source datasets. The data in those datasets have been collected in diverse environments (both indoor and outdoor).

Chapter 2

Preliminaries and Methodology

This section presents the mathematical notations used in this paper, along with the basics of graph signal processing. Further, we discuss the framework to recover missing data in sensor networks which is inspired by the theory of reconstruction of time-varying graph signals [18, 21] by minimizing the Sobolev norm [17].

2.1 Mathematical Notations

For the convenience of readers, firstly we introduce the various notations used in this paper. The uppercase boldface letters like $\mathbf{X}$ represents the matrices, lowercase boldface letters like $\mathbf{x}$ denote vectors. Calligraphic letters like $\mathcal{L}$ denotes sets. Matrix products like Hadamard and Kronecker products are denoted by $\circ$ and $\otimes$, respectively. The vectorization of $\mathbf{X}$ is denoted by $\text{vec}(\mathbf{X})$. $(\mathbf{X})^T$ denotes the transpose of matrix $\mathbf{X}$. The diagonal of a matrix with entries $x_1, x_2, x_3, \dots, x_n$ is denoted by $\text{diag}(\mathbf{x})$. The ℓ_2 and Frobenius norm of a vector $\mathbf{x}$ are represented by $\|\mathbf{x}\|_2$, $\|\mathbf{x}\|_F$. The trace of a matrix $\mathbf{X}$ is given by $\text{tr}(\mathbf{X})$. Any other mathematical symbol used in this paper bear their usual significance.

2.2 Graph Construction and Graph Signals Representation

Let us consider an undirected weighted graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{W})$. Here $\mathcal{V}$ is the set of N nodes/vertices with $|\mathcal{V}| = N$. $\mathcal{E} = \{(i, j)\}$ represents the set of edges with (i, j) as an edge between the nodes i and j . $\mathbf{W}$ is the weighted adjacency matrix.

According to the spectral graph theory [3] and graph signal processing [b11], we define

the unnormalized Laplacian of G as $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the diagonal matrix of G and $\mathbf{D}(i, i) = \sum_{j=1}^N \mathbf{W}(i, j) \forall i = 1, \dots, N$. The eigenvalues of $\mathbf{L}$ are $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ and their corresponding eigenvectors are $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_N\}$.

Finally, on each node, we define a graph signal that is a function on v such that $x : v \rightarrow \mathbb{R}$. The graph signal can be represented as a vector $\mathbf{x} \in \mathbb{R}^N$, where $\mathbf{x}(i)$ is the value of the function in the node $i \in v$.

2.3 Reconstructing Time-varying Graph Signals by Sobolev Norm Minimization

In the theory of graph signal processing, sampling and reconstructing graph signals play an essential role [7]. For reconstructing a graph signal from its samples, the graph signals need to be bandlimited [16]. In the literature of GSP, most of the proposed recovery algorithms make a prior assumption that graph signals are smooth in graphs. In our work involving sensor networks, we also consider graph signals to be spatiotemporally smooth. This is because adjacent sensor readings are similar in nearby positions and the readings are progressive with time. For static graph signals in G , the graph Laplacian quadratic form $S_2(\mathbf{x}) = \mathbf{x}^T \mathbf{L} \mathbf{x}$ is widely used as a measure of smoothness[22]. However, in sensor networks, missing data may arise at any point in space-time; hence we need to consider both the spatial and the temporal distribution of missing data while we try to reconstruct their original values. So the concept of time-varying graph signals is ideal for dealing with these forms of missing data. Qiu *et al.* [18] extended the definition of $S_2(\mathbf{x})$ to time-varying graph signals. Let a time-varying graph signal be expressed as a matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M]^T$, where $\mathbf{x}_t$ denotes the signal at time t ($1 < t < M$) in G . Here each row of $\mathbf{X}$ represents a time-series on the corresponding vertex. Now we define the smoothness of $\mathbf{X}$ as [18]:

$$S_2(\mathbf{X}) = \sum_{t=1}^M S_2(\mathbf{x}_t) = \text{tr}(\mathbf{X}^T \mathbf{L} \mathbf{X}) \quad (2.1)$$

As per Qiu *et al.*, the temporal difference operator matrix $\mathbf{D}_h \in \mathbb{R}^{M \times (M-1)}$ is defined as:

2.3 Reconstructing Time-varying Graph Signals by Sobolev Norm Minimization

$$\mathbf{D}_h = \begin{bmatrix} -1 & & & & \\ & 1 & -1 & & \\ & & 1 & \ddots & \\ & & & \ddots & -1 \\ & & & & 1 \end{bmatrix} \in \mathbb{R}^{M \times (M-1)}, \quad (2.2)$$

This is done so as to include temporal information in the problem of time-varying graph signal reconstruction. Also, the temporal difference signal is defined as:

$$\mathbf{X}\mathbf{D}_h = [\mathbf{x}_2 - \mathbf{x}_1, \mathbf{x}_3 - \mathbf{x}_2, \dots, \mathbf{x}_M - \mathbf{x}_{M-1}]. \quad (2.3)$$

In their work, Qiu *et al.* had proposed two different reconstruction methods, one for the noisy case and the other for the noiseless case. Since sensor networks accumulate sufficient noise during their operational period, we consider the noisy case in our work. As per Qiu *et al.*, the noisy case is defined as follows:

$$\min_{\bar{\mathbf{X}}} \frac{1}{2} \|\mathbf{J} \circ \bar{\mathbf{X}} - \mathbf{Y}\|_F^2 + \frac{\gamma}{2} \text{tr}((\bar{\mathbf{X}}\mathbf{D}_h)^T \mathbf{L} \bar{\mathbf{X}}\mathbf{D}_h), \quad (2.4)$$

Here $\mathbf{Y}$ is the sampled matrix and the sampling matrix for the whole time-varying graph signal is $\mathbf{J} \in \{0, 1\}^{N \times M}$, defined as:

$$\mathbf{J}(i, j) = \begin{cases} 1 & \text{if } i \in S_t, \\ 0 & \text{if } i \notin S_t, \end{cases} \quad (2.5)$$

S_t being the sampled set of vertices at time t . In a nutshell, Eqn. 2.4 reconstructs a time-varying graph signal $\bar{\mathbf{X}}$ with a small error $\|\mathbf{J} \circ \bar{\mathbf{X}} - \mathbf{Y}\|_F^2$ while minimizing the temporal difference graph signal smoothness $\text{tr}((\bar{\mathbf{X}}\mathbf{D}_h)^T \mathbf{L} \bar{\mathbf{X}}\mathbf{D}_h)$. γ in Eqn. 2.4 is known as the regularization parameter and it weights the importance between the error and smoothness terms.

In 2020, Giraldo and Bouwmans [17] proposed a new algorithm for time-varying graph signals reconstruction inspired by the minimization of the Sobolev norm. The norm was first defined by Pesenson *et al.* [23] for introducing the variational problem in graphs as:

Sobolev norm: Let $\mathbf{L}$ and $\mathbf{x}$ be the Laplacian matrix and graph signal, respectively. Let

2.3 Reconstructing Time-varying Graph Signals by Sobolev Norm Minimization

$\varepsilon \geq 0$ and $\beta \in \mathbb{R}^+$ be two constant parameters, and the Sobolev norm is defined as follows:

$$\|\mathbf{x}\|_{\beta, \varepsilon} = \|(\mathbf{L} + \varepsilon \mathbf{I})^{\beta/2} \mathbf{x}\| \quad (2.6)$$

When $\mathbf{L}$ is symmetric, we have that as:

$$\mathbf{x}^T (\mathbf{L} + \varepsilon \mathbf{I})^{\beta} \mathbf{x} \quad (2.7)$$

In their work, they found that the term $(\mathbf{L} + \varepsilon \mathbf{I})$ in Eqn. 7 has a better condition number than $\mathbf{L}$ when $\varepsilon > 0$. By extending the concept of Sobolev-norm to time-varying graph signals (considering $\mathbf{L}$ to be a symmetric matrix), we get:

$$\|\mathbf{X}\|_{\beta, \varepsilon} = \sum_{i=1}^M \mathbf{x}_i^T (\mathbf{L} + \varepsilon \mathbf{I})^{\beta} \mathbf{x}_i = \text{tr}(\mathbf{X}^T (\mathbf{L} + \varepsilon \mathbf{I})^{\beta} \mathbf{X}) \quad (2.8)$$

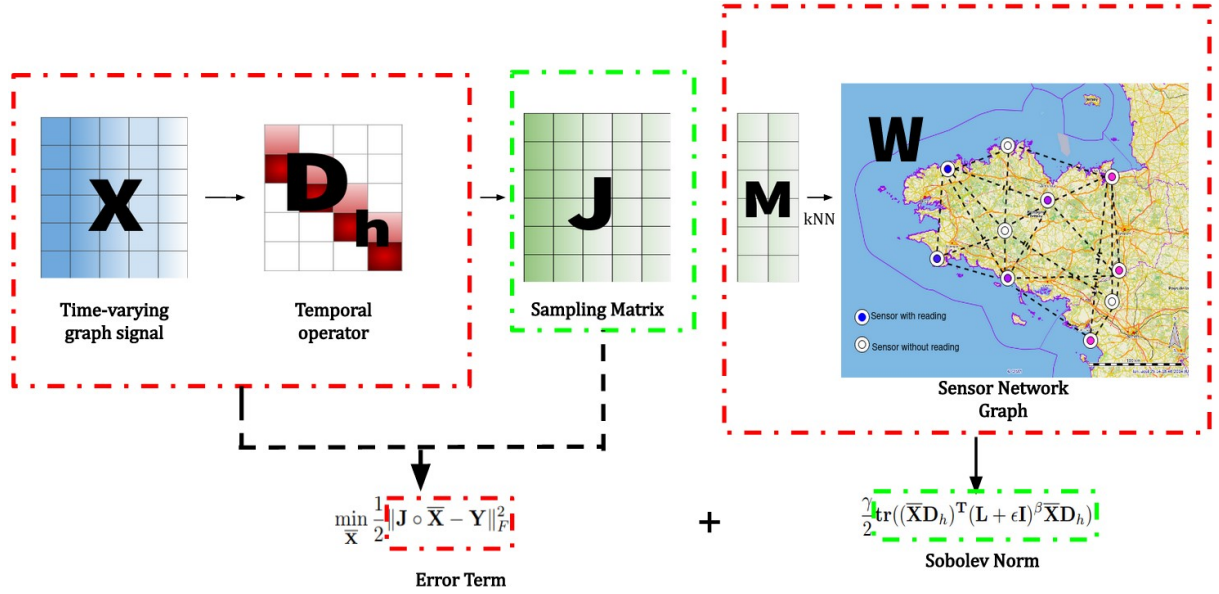


Figure 2.1 Schematic diagram of the proposed method.

Finally, the Sobolev reconstruction problem for time-varying graph signals is formulated as:

$$\min_{\bar{\mathbf{X}}} \frac{1}{2} \|\mathbf{J} \circ \bar{\mathbf{X}} - \mathbf{Y}\|_F^2 + \frac{\gamma}{2} \text{tr}((\bar{\mathbf{X}} \mathbf{D}_h)^T (\mathbf{L} + \varepsilon \mathbf{I})^{\beta} \bar{\mathbf{X}} \mathbf{D}_h) \quad (2.9)$$

Here we use the temporal difference operator of Eqn. 2.2 along with the Sobolev-norm of time-varying graph signals of Eqn. 2.8. The schematic diagram of our method is shown in Fig. 2.1.

Chapter 3

Experimental Framework and Evaluation

This section briefly discusses the implementation of our proposed method and the dataset used to assess it. We also discuss the evaluation metrics and compare our method with SoTA baselines [4, 12, 13, 20] on those metrics.

All the experiments were performed in MATLAB® 2018a on a Intel® Core™ i5 8th Gen. Laptop with Linux Mint and having 4 CPUs with a clock frequency of 1.6 GHz each. We used Python 3.10.0 for generating the figures and graphs in the paper.

3.1 Datasets

To evaluate the performance of our proposed method, we test it on many publicly available widely used sensor-network datasets. To test the flexibility of the proposed method, we use datasets collected in diverse environments (indoor and outdoor).

Molène Dataset: [24] The French national meteorological service¹ published an open-access dataset of hourly weather observations in Brittany, France, for the month of January 2014. In addition to the graph of ground weather stations, the dataset contains hourly readings of those stations. Readings include temperatures, wind characteristics, rain, and other information.

Intel Lab Dataset: [25] This dataset contains data collected from 54 sensors deployed in the Intel Berkeley Research lab between February 28th and April 5th, 2004². Sensors measured various parameters like temperature, humidity, light intensity, and voltage values at an interval

¹<http://data.gouv.fr>

²<http://db.csail.mit.edu/labdata/labdata.html>

of 30 seconds in an indoor environment. For convenience in processing and representing the data, we consider the temperature readings only.

3.2 Evaluation Metrics

To compare our method against baselines, we use two of the most widely used metrics in sensor networks, namely the **Root Mean Square Error (RMSE)** and the **Mean Absolute Error (MAE)**. The two metrics are defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - x_i^*)^2} \quad (3.1)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - x_i^*| \quad (3.2)$$

Here n is the number of unavailable/ missing readings, x_i is the ground truth (original) data and x_i^* is the reconstructed/ recovered data using the proposed method.

3.3 Experiments and Results

Let $\mathbf{M} \in \mathbb{R}^{N \times 2}$ be the matrix of positions of all sensor nodes in $\mathcal{V}$ such that $\mathbf{M} = [\mathbf{m}_1, \dots, \mathbf{m}_N]^T$, where $\mathbf{m}_i \in \mathbb{R}^2$ is the vector with the latitude and longitude (for the Moléne dataset) or x and y coordinates (for the Intel dataset) of vertex i . To connect the vertices of the graph, we use a k -nearest neighbors (kNN) strategy. The value of k is determined experimentally. The weights of each edge (i, j) connecting the nodes i and j is given by $\mathbf{W}(i, j) = \exp(-\frac{\delta(i, j)^2}{\sigma^2})$, where $\delta(i, j) = \|\mathbf{m}_i - \mathbf{m}_j\|_2$ is the Euclidean distance between the nodes i and j and σ^2 is the standard deviation of the Gaussian kernel $\sigma = \frac{1}{|\mathcal{E}|} \sum_{(i, j) \in \mathcal{E}} \delta(i, j)$.

We randomly remove some data points from the dataset and then recover them to evaluate the Sobolev reconstruction model's performance. For a fair comparison, we test the baseline methods on the same datasets on which we test our method. We also scale the data points in the two datasets to the range $[0, 1]$. The reconstructed values are scaled to this range as well. For the sampling matrix $\mathbf{J}$, we take a random sampling strategy so that each time-graph signal $\mathbf{x}_t$ has the same number of sampled nodes $\forall t$ ($1 < t < M$). Also, we empirically tune the values of constants ε , γ , and β to get the combination of values for which the performance of

Table 3.1: Performance of the proposed method under different sampling densities in the two datasets

Sampling Density	Molène Dataset		Intel Dataset	
	RMSE	MAE	RMSE	MAE
0.1	2.40	1.65	3.04	1.50
0.3	1.60	1.02	2.01	0.97
0.5	1.20	0.74	1.57	0.70
0.7	0.97	0.65	1.30	0.55

our proposed method is the best. Again, for fairly comparing the studied algorithm with the baselines (kNN, Expectation-Maximization, LRMC, and PMF), we find out the best values of different parameters affecting their performances.

For the Molène dataset, the value of nearest-neighbors is determined experimentally and is set to $k = 5$. Since all the weather stations do not have readings over the whole period of 31 days, we consider the temperature readings of 10 such stations where the readings are consistently available. From the 744 hours of readings, we randomly remove some readings while using the available data from the rest of the dataset. The sampling densities are set to $\{0.1, 0.3, 0.5, 0.7\}$, where each sampling density means the amount of valid information. For example, when the sampling density is 0.1 the reconstructed data is 90% and the available data is 10%.

For the Intel dataset, the value of k is set to 3. Here we consider the first 10^4 readings out of 10^5 readings ($38 \text{ days} \times 24 \text{ hours} \times 60 \text{ minutes} \times 2$). In this dataset too, we randomly remove some values with sampling densities $\{0.1, 0.3, 0.5, 0.7\}$ and reconstruct those values using our method.

For both datasets, we perform Monte-Carlo cross-validation [26] with 20 repetitions. Finally we compare the reconstructed values with the ground truth data and evaluate the performance of our algorithm.

Table 3.1 shows the performance of our algorithm under different sampling densities. Here we observe that our method performs well under higher sampling densities than under lower ones. Since the number of available readings increases with increasing sampling density, we are getting richer information about the underlying spatio-temporal information in the dataset. So we are getting better performance while recovering the missing values. In Table 3.2, we compare the proposed method against the baselines [4, 12, 13, 20] with different sampling densities. Fig 3.1 shows that even though other methods fail to recover the missing values under

Table 3.2: Quantitative comparison results of SoTA methods vs proposed method. Best results are in **bold**.

	Sampling Density	kNN [12]		EM [13]		LRMC [20]		PMF [4]		Proposed	
		RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Molene Dataset	0.1	8.05	6.75	7.34	5.68	3.47	2.35	4.75	2.56	2.40	1.65
	0.3	5.45	4.68	5.20	3.93	2.82	1.35	3.42	1.64	1.60	1.02
	0.5	4.20	3.04	4.14	2.95	1.96	1.02	2.45	1.09	1.20	0.74
	0.7	3.40	2.13	3.19	1.82	1.02	0.81	1.22	0.95	0.97	0.65
Intel Dataset	0.1	9.70	5.34	7.78	5.63	4.68	2.94	5.34	3.98	3.04	1.50
	0.3	6.30	3.52	6.45	4.96	3.40	2.15	3.95	2.37	2.01	0.97
	0.5	4.60	2.98	5.17	3.81	2.06	1.21	2.75	1.65	1.57	0.70
	0.7	3.75	1.78	4.63	2.57	1.45	0.75	1.97	0.99	1.30	0.55

lower sampling densities accurately, our method significantly outperforms them by providing consistent performance under those conditions. One explanation behind our model’s success can be its ability to capture the spatial and temporal correlations inside the data effectively. Moreover, there is no prior knowledge about the behavior of the underlying processes in the time domain affecting the sensor readings. Even then, our method successfully recovers the missing values.

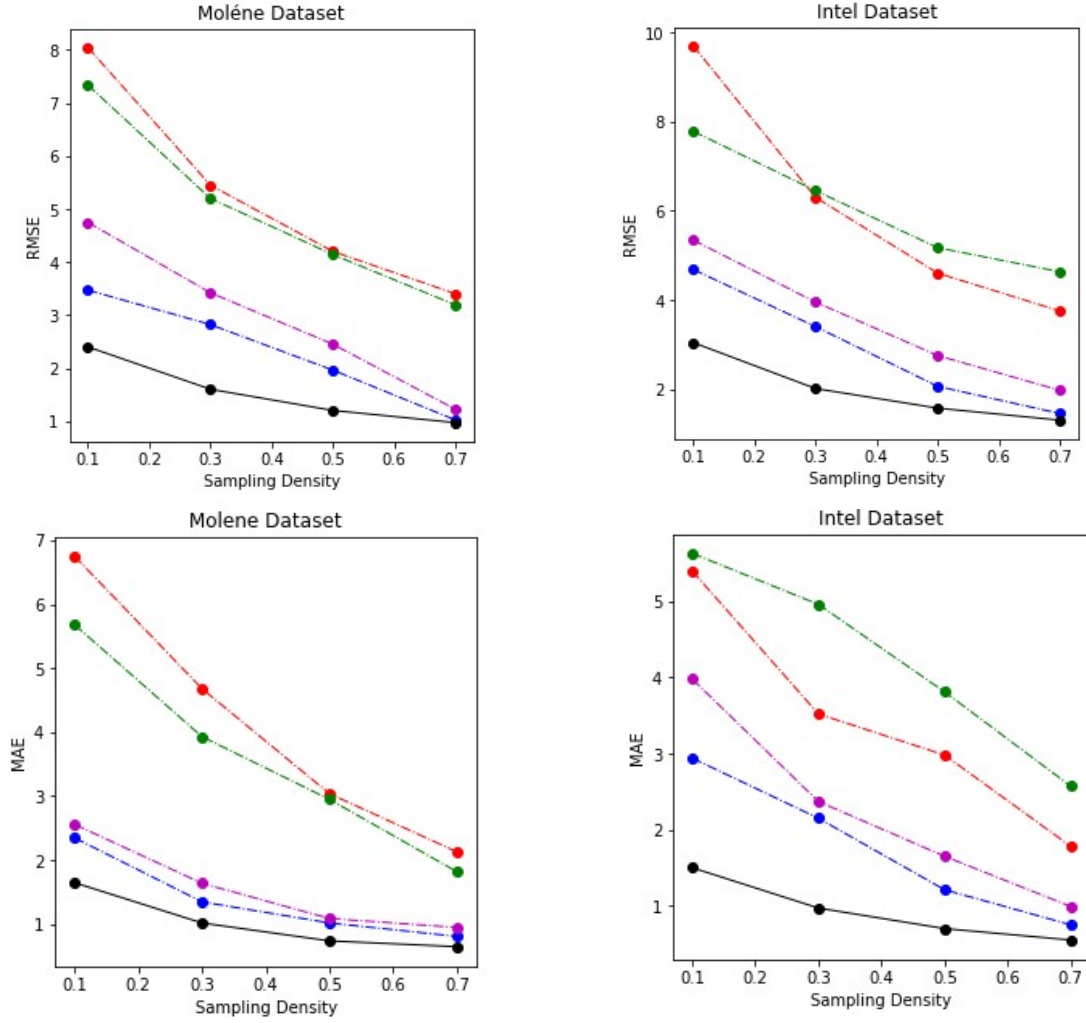


Figure 3.1 Performance of various SoTA algorithms in recovering missing sensor readings. kNN is marked with Red, EM is marked with Green, LRMC is marked with Blue, PMF is marked with Purple, and the proposed method is marked with Black.

Chapter 4

Conclusions and Future Directions

In this paper, we introduce a time-varying graph signals reconstruction approach to solve the problem of missing sensor data recovery in wireless sensor networks. We show how the proposed method performs well under massive data loss situations, even without prior knowledge of the factors affecting the sensor readings. We also validate the proposed method on several publicly available datasets belonging to diverse environmental conditions to demonstrate its flexibility. We believe that our approach will help future researchers to use the promising field of GSP in solving various related problems like intelligent transportation systems, weather conditions forecasting, disaster management, and others.

We also plan to extend the applications of the proposed algorithm to non-smooth situations, for example, where nearby localities have dissimilar readings (e.g., in mountainous and plateau regions) and changes in weather conditions are abrupt with time. Using graph learning to perform the recovery task can also be a way forward.

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The findings of this research are accepted for publication at the **European Signal Processing Conference (EUSIPCO) 2022**, to be held between **29 August 2022 – 2 September 2022** in **Belgrade, Serbia**.

Paper title: *Recovery of Missing Sensor Data by Reconstructing Time-varying Graph Signals*

Authors: Anindya Mondal, Mayukhmali Das, Aditi Chatterjee, Palaniandavar Venkateswaran

The preprint is available at <https://arxiv.org/abs/2203.00418>.

EUSIPCO 2022 Review Results [Paper #1590]

EUSIPCO 2022 <eusipco2022@cmsworkshops.com>

Mon, May 16, 2022 at 9:25 PM

To: Anindya Mondal <anindyam.jan@gmail.com>, Mayukhmali Das <mayukhmalidas322@gmail.com>, Aditi Chatterjee <eice.aditi@gmail.com>, Palaniandavar Venkateswaran <p.venkateswaran@jadavpuruniversity.in>

Dear Anindya Mondal, Mayukhmali Das, Aditi Chatterjee, Palaniandavar Venkateswaran,

Paper ID: 1590

Title: RECOVERY OF MISSING SENSOR DATA BY RECONSTRUCTING TIME-VARYING GRAPH SIGNALS

The review process for the 30th European Signal Processing Conference (EUSIPCO 2022) has now been finalized, and it is our pleasure to inform you that your paper has been accepted for publication. Congratulations! The review comments that have led to this decision can be found below.

Please note that due to the COVID-19 pandemic, which is not over yet, the event will be hybrid, that is, both physical and virtual presence will be possible. Of course, we do hope that the physical part will be the dominant one, and you will be able to present your paper in person and gain the full experience of a physical conference.

Carefully consider the following instructions for the publication of your paper and your participation in the conference:

- A camera-ready PDF version of your paper needs to be submitted by June 12, 2022 in the online submission system: <https://cmsworkshops.com/EUSIPCO2022/Papers.asp>. Please make sure to take into account the below review comments when preparing your camera-ready paper. The layout of the camera-ready paper is identical to the layout of the submitted paper. Details can be found in the Author's Information page on the EUSIPCO 2022 website. Please do not include page numbers or headers/footers in your camera-ready paper.

- Your paper needs to be covered by a regular (non-student) conference registration by June 14, 2022. This requirement also holds for papers authored or presented by students. Registration will soon open on the conference website: <https://2022.eusipco.org/>. When registering, you will be asked to enter the Paper ID of the papers you wish to cover with your registration. Papers that are not covered by a regular (non-student) conference registration by June 14, 2022 will be removed from the conference proceedings. Detailed instructions will be soon available on the conference website.

- In the following days, with another email, you will be informed about the format of your paper (lecture/poster) and on its location in the conference program. You will be asked to record a video for presenting your work, and we will provide you with a link for the instructions.

- The conference proceedings will be published by September 2022 in the open-access proceedings archive of the European Association for Signal Processing (EURASIP) and on IEEE Xplore.

We look forward to meeting you (physically or virtually) in Belgrade in August 2022!

Sincerely,

Aleksandra Pizurica, Kostas Berberidis, Pau Closas
EUSIPCO 2022 Technical Program Co-Chairs

---- Comments from the Reviewers ----

Review #024F

Paper Format: Correctly formatted

Topic relevance: High

Originality of the content: Low

Methodology / Research design: Low

Evaluation of results and derived conclusions: Average

References to previous work: High

Correct English usage: Average

General Comments to Authors

Please include the following content in a .tex file and compile it.

```
\documentclass[a4paper,11pt]{article}
\usepackage[utf8]{inputenc}
\usepackage{geometry}
\usepackage{latexsym}
\usepackage{amsmath,amssymb,amsfonts}
\usepackage{amssbsy}
\usepackage{bm}
\usepackage{color}
\usepackage{mdframed}
```

```
\begin{document}
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```
\begin{center}
```

```
\large Review of paper ``Recovery of Missing Sensor Data by Reconstructing Time-varying Graph Signals"
\end{center}
```

This paper deals with the problem of missing data in wireless sensor networks seen as time-varying graph signals. In particular, authors extend the work of [14] by involving the Sobolev norm in the reconstruction of incomplete signals.

My first concern is the lack of description of your method. I understand that (9) is $\textit{your}$ minimization problem, but I first thought that it was part of the work of [12]. A lot of space is dedicated to describe the methods on which your approach is based, *i.e.*, [12] and [14] mainly.

Secondly, the main methodological contribution is that authors replace the graph Laplacian $\bm{L}$ in the minimization problem (4) by $(\bm{L} + \epsilon \bm{I})^\beta$, which is a regularized version of the former with better conditioning. In essence, the minimization problem of [14] is replaced with the graph signal smoothness difference of [12]. So in my opinion, the amount of novelty is fairly small.

A third important concern is the comparison to other methods in numerical experiments. Because the methodological framework is very similar to the one in [14], I think a comparison with [14] should be considered to evaluate the main extension of the present study, *i.e.*, the change from $\bm{L}$ to $(\bm{L} + \epsilon \bm{I})^\beta$ in the reconstruction scheme. Otherwise I found the experiments well designed, with an interesting empirical property that the error variation is independent from the type of considered metric.

Finally, authors do not mention the underlying missing-data mechanism, which is encoded in the binary mask or sampling matrix $\bm{J}$. Therefore I assume that the setting is the missing completely at random (MCAR) one, which says that the conditional probability of $\bm{J}$ given $\bm{X}$ is independent of the data. Why did you choose such a setting? Was it by default? In your opinion, can your method handle missing not at random (MNAR) or missing at random (MAR) data? If not, could it be a perspective of your work?

Other remarks:

```
\begin{itemize}
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\item[--] Please consider using command \texttt{\textbackslash eqno} to cite equations.
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\item[--] Please add punctuation to your equations.
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\item[--] There are some typos and incorrect formulations.
```

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\end{itemize}
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\end{document}
```

Review #0FEA

Paper Format: Correctly formatted

Topic relevance: Very High

Originality of the content: Very High

Methodology / Research design: Average

Evaluation of results and derived conclusions: Very High

References to previous work: Average

Correct English usage: Very High

General Comments to Authors

The authors present a method for recovery of missing sensor data. They follow a computer science approach, i.e., not really looking at the physical interpretation of the data. I would expect that much better results could be obtained taking a physical model of the data into account.

Review #03E6

Paper Format: Minor improvements needed

Topic relevance: Very High

Originality of the content: Very High

Methodology / Research design: High

Evaluation of results and derived conclusions: Very High

References to previous work: Very High

Correct English usage: High

General Comments to Authors

In this paper, a time-varying graph signals reconstruction approach to solve the problem of missing sensor data recovery in wireless sensor networks is proposed. The proposed approach outperforms state-of-the-art algorithms such as the PMF and LRMC. The authors use datasets to validate their results.

-end-